

IMAGE PROCESSING TECHNIQUES

Extended Class Activities Worksheet | UG Level | Activity 7: Edge Detection Techniques

PART 1: EDGE DETECTION TECHNIQUES COMPARISON

Operator	Type (Derivative Order / Category)	Noise Sensitivity
Sobel	First-order Derivative (Gradient-based)	Moderate (smoothed by weighted kernel coefficients)
Prewitt	First-order Derivative (Gradient-based)	Moderate (unweighted averaging smooths noise)
Canny	Multi-stage Optimal Edge Detector	Low (uses Gaussian filtering step first)
Roberts	First-order Derivative (2x2 Diagonal Gradient)	High (highly sensitive due to small 2x2 kernel size)

PART 2: WHY CANNY IS CALLED AN OPTIMAL EDGE DETECTOR

John F. Canny mathematically defined edge detection as an optimization problem based on three essential performance criteria:

- **1. Low Error Rate (Good Detection):** The algorithm maximizes the detection of true edge points while minimizing false edges caused by noise.
- **2. Good Localization:** The detected edge pixels are located as close as possible to the actual center of the real physical edge in the image.
- **3. Single Response Constraint:** The detector produces only one single response (single-pixel-wide thin line) for each true edge, avoiding thick or duplicate boundary lines via non-maximum suppression.